MIL-C-18539(NOrd)

5 May 1955

SUPERSEDING
NAVORD OS 3133H
28 October 1953

MILITARY SPECIFICATION

CARTRIDGE, SLICK MARKER,

AN-MK 1 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of the Cartridge, Slick Marker AN-Mk 1 Mod 0 and the methods of inspection and tests upon which acceptance will be based. The marker consists of a shell assembly, housing a fluorescein dye container which is assembled with a delay fuze and bursting charge. The cartridge, when fired from a pistol, propels the dye container and ignites the delay fuze, which in turn, initiates the bursting charge. The dye is expelled and a distinctive color is produced on the surface of the water.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-601 -- Boxes; Wood, Cleated-Plywood.

MILITARY

JAN-P-106 -- Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.

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JAN-P-108 -- Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-board and W-board), Exterior and Interior.

JAN-Q-378 -- Quickmatch.

MIL-V-16399 -- Varnish. Moistureproof.

STANDARDS

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 -- Marking for Shipment and Storage

DRAWINGS

BUREAU OF ORDNANCE

LD SK 102963 -- Cartridge, Slick Marker AN-Mk 1 Mod 0, and all drawings, standards, and specifications listed thereon.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-

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Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General.

3.1.1 The Cartridge, Slick Marker AN-Mk 1 Mod 0 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD SK No. 102963 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Materials.

3.2.1 Primer. -- The primer shall be a commercial grade, Winchester No. 4, or equal.

3.2.2 Delay fuze. -- The time fuze shall be a high grade commercial waterproofed fuze, taped and double countered. The burning rate shall be 45 ± 5 seconds per foot. The quality shall be comparable to the Ensign Bickford Company's "Crescent" brand, or equal.

3.2.3 Quickmatch. -- The quickmatch shall conform to the requirements of Specification JAN-Q-378, Type I, Class B.

3.2.4 Dye material.

3.2.4.1 Composition. -- The dye material shall contain 86 percent minimum of soluble sodium salt of fluorescein, 14 percent maximum of other water soluble salt, and 0.50 percent maximum of insoluble material. The dye material shall pass the tests specified in 4.5.1.1 and 4.5.1.3.

3.2.4.2 Purity. -- The dye shall contain no foreign materials which will in any way delete from its brilliance and solubility. The dye shall not contain any de-dusting agent.

3.2.4.3 Color. -- A salt water solution of the dye shall produce a yellow-green color.

3.2.4.4 Moisture content. -- The dye powder shall have a maximum moisture content of 10 percent so that clumping will

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not result after it has been packed in the container. The moisture content shall be determined by the test specified in 4.5.1.2.

3.2.4.5 Fineness. -- The size of the dye particles shall be such that all will pass through a U. S. Standard No. 40 mesh sieve.

3.3 Coating.-- The completed assembly shall be dipped or sprayed, with moistureproof varnish conforming to the requirements of Specification MIL-V-16399. The varnish shall not be thinned with more than 25 percent varnish thinner.

3.4 Functioning. -- The cartridges shall pass the functioning test specified in 4.5.2. Cartridges shall effectively disperse the dye to produce a serviceable slick.

3.5 Delay time. -- The cartridges shall pass the delay time test specified in 4.5.3. Cartridges shall have a delay time of 11 ± 1 seconds.

3.6 Assembly. -- The cartridges shall be assembled in accordance with the requirements of the applicable drawings and as specified in 3.6.1 through 3.6.3.

3.6.1 Shell.

3.6.1.1 Insert the rocket paper shell body into the brass base.

3.6.1.2 Place the unpressed wad into the liner, and insert the unit into the paper down upon the base.

3.6.1.3 With the base in a suitable holder, press the wad under 500 pounds pressure using an appropriate ram.

3.6.1.4 Roll crimp the brass base onto the tube following standard crimping practice.

3.6.2 Dye container.

3.6.2.1 Insert the safety fuze into the tubular adapter and crimp in place.

3.6.2.2 Assemble the felt and cardboard washers over the fuze adapter down onto the flange.

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3.6.2.3 Assemble one metal washer over the safety fuze at the flanged end, and slit this end of the fuze.

3.6.2.4 Insert the quickmatch and secure with thread.

3.6.2.5 Load the required bursting charge into the dye container, and insert the fuze assembly into the dye container over the bursting charge.

3.6.2.6 Load the required amount of fluorescein dye and cover with the cork washer.

3.6.2.7 Over the cork washer, position the container cover, and seal with a suitable lacquer.

3.6.2.8 Slit the protruding end of the time fuze, insert the quickmatch and secure as before.

3.6.3 Final assembly.

3.6.3.1 Load the required propelling charge into the shell, filling the cavity in the shell wad. Secure a cloth disc to the felt washer before inserting into the shell.

3.6.3.2 Insert the assembled dye container, and place cardboard closing discs over the dye container.

3.6.3.3 Crimp the end of the cartridge to securely hold the contents in place.

3.7 Workmanship. -- Only the best trade practices shall be utilized in the manufacture of the Cartridge, Slick Marker AN-Mk 1 Mod 0. Care shall be exercised in the assembly of the component parts to assure proper seating and alignment especially where crimping is employed to maintain parts in their relative positions. There shall be no burrs or other imperfections of manufacture harmfully affecting the functioning or assembly of the product. All parts shall be free of chips, dirt and other foreign materials.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

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4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of twenty-five (25) Cartridges, Slick Markers AN-Mk 1 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These cartridges shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For procurement purposes the production lot shall consist of five hundred (500) cartridges. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be provided at the expense of the Contractor. The production lot should not be confused with the lot for inspection purposes as defined in Standard MIL-STD-105.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

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4.5 Test procedures.

4.5.1 Dye material test.

4.5.1.1 Sodium fluorescein content.

4.5.1.1.1 Pulverize and dry the fluorescein dye material, as received, to constant weight on a watch glass at 105 degrees C.

4.5.1.1.2 Place the dye in a weighing bottle and again dry to constant weight at 105 degrees C.

4.5.1.1.3 Using a weighing bottle to sample the dye, accurately weigh a one gram sample of dye and dissolve in a beaker containing 200 milliliters of distilled water.

4.5.1.1.4 Heat the solution to boiling, and add 10 percent hydrochloric acid a few drops at a time with constant stirring until the solution gives a distinct acid end-point with congo-red paper. (The acid must be added while the solution is boiling to prevent the formation of a more soluble form of yellow fluorescein.)

4.5.1.1.5 Allow the solution to cool to room temperature and filter through a tared gooch crucible using 75 milliliters of distilled water to wash the precipitate into the crucible.

4.5.1.1.6 Dry for one hour in an oven at 100 to 105 degrees C.

4.5.1.1.7 The percentage of soluble fluorescein salt is equal to the weight of precipitate multiplied by 113.

4.5.1.2 Moisture content.

4.5.1.2.1 Accurately weigh a two gram sample of dry material into a weighing bottle previously brought to constant weight at 105 degrees C.

4.5.1.2.2 With the bottle open, dry it to constant weight at 105 degrees C, close, and weigh again.

4.5.1.2.3 Weight lost divided by weight of original sample multiplied by 100 gives the percentage of moisture content.

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4.5.1.3 Insoluble material content.

4.5.1.3.1 Dissolve 30 ± 1 grams of dye material (as calculated on a moistureproof basis) in 100 milliliters of hot water.

4.5.1.3.2 Gently boil the solution for 15 minutes keeping the water level constant; then filter, while hot, through a weighed gooch crucible and wash.

4.5.1.3.3 Dry the crucible to constant weight at 105 degrees C.

4.5.1.3.4 The increase in weight (in grams) divided by the weight of sample on a dry basis multiplied by 100 gives the percentage of total insoluble contents of dry material.

4.5.1.3.5 The percentage of insoluble contents of dry material shall not exceed 0.50 percent.

4.5.2 Functioning test. -- Select and test samples from each production lot in accordance with Double Sampling Plan, AQL 2.5 percent, Table G of Standard MIL-STD-105. The cartridges shall be fired from an AN-M8 pyrotechnic pistol carried in an airplane at an altitude between 200 and 300 feet travelling at a speed of approximately 150 knots. The pistol shall be aimed downward and aft at an angle of approximately 45 degrees. Failure of the cartridges to properly expel the dye container or to ignite the delay fuze, and failure of the bursting charge to thoroughly disperse the dye, thereby producing no serviceable slick, shall be cause for rejection.

4.5.3 Delay time test. -- Stop watch measurements of delay time shall be recorded for a sufficient number of the cartridges subjected to the functioning test of 4.5.2 to insure conformance with Single Sampling Plan, AQL 6.5 percent, Table G of Standard MIL-STD-105. Cartridges shall meet a delay time of 11 ± 1 seconds.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -- Ten (10) markers shall be packaged in a Style RSC fiberboard carton in accordance with Specification JAN-P-108, compliance symbol W5c. Partitions and separators shall conform to the same specification.

5.2 Packing. -- Two hundred (200) markers packaged as described in 5.1 shall be packed in a wood-cleated plywood

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or nailed wood box conforming to Specification PPP-B-601 or Specification JAN-P-106.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 71-78.

5.4 Data cards.

5.4.1 Manufacturing.

5.4.1.1 For Department of the Navy purchases three copies of Pyrotechnic Manufacturing Data Cards (NAVORD Form 1204) shall be completed by the manufacturer for each lot.

5.4.1.2 The Government Inspector shall complete the acceptance data section (NAVORD Form 1204) by listing the following components with data called for:

Primer
Delay fuze
Quickmatch
Fluorescein dye

The Government Inspector shall then endorse the cards and forward them to the Bureau of Ordnance.

5.4.2 Performance.

5.4.2.1 For Department of the Navy purchases one Pyrotechnic Ammunition Performance Data Card (NAVORD Form 1187) shall be completed, and enclosed, one for each 50 markers, as specified in 5.2 in such a manner that the form will reach the point of use of the cartridges.

5.4.2.2 The description shall be indicated on the first two top lines of the form front.

5.4.2.3 The references shall be: NAVORD OCL No. AV6-44.

5.4.2.4 The performance elements shall be:

Primer
Propulsion
Delay fuze (11 ± 1 seconds)
Bursting action
Spread of dye

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6. NOTES

6.1 Ordering data. -- Procurement documents should specify the title, symbol and date of this specification.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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